An Index of CLIP Read Architectures
Compiled manually by Joseph Luna, May-August 2026 v2
Visit: https://github.com/LunaRNALab/CLIPittyClip for the most up to date version of this guide

Key: UMI (Unique molecular identifier per sequence)
BC (Barcode, unique per library, sometimes called an index)
Spacer (sequence that serves as a spacer, usually for ligation bias mitigation purposes)
Anchor (a fixed nucleotide that can be used to register component starts, also acts as a spacer)
READ (your read of interest)
3' adapter (what is immediately downstream of your read, usually used as an RT handle, but not always)

The goal of this guide is to provide an at-a-glance understanding of the major CLIP read architectures out there, and to see how they are handled by CLIPittyClip. Our own coCLIP implementation stems from the BrdU antibody pulldown variant of iCLIP, so we'll start there. Note that for all versions BC length is auto computed if a -b barcodes.txt file is supplied. Otherwise, specify BC length can be specified with --bc-len.

1) coCLIP read architecture Current Luna lab standard as published in Yi et al RNA 2024, derived from Darnell and Zhang labs BrdU-CLIP protocols that were first published in the 2014 Cell Reports paper (PMID: 24613350). The RT primers in our implementation were first described in Sawicka et al 2019 eLife (PMID: 31860442).

5'NNNNNNN**XXXXXX**DDDREADGTGTCAGTCACTTCCAGCGG'3
UMI BC sp L32 adapter

CLIPittyClip args:
-u 7
--bc-len 6
--spacer-len 3
-a GTGTCAGTCACTTCCAGCGG ← L32 default, no need to specify

2) BrdU-CLIP read architecture From Weyn-Vanhentenryck et al 2014 Cell Rep PMID: 24613350. Note that primers initially had 4mer barcodes, plus an anchor position C at either end of the UMI, and that the orientation is reversed (BC then UMI). Here we will treat first anchor as part of the BC and the second anchor as a spacer

```
       C        C Anchor positions, treated as BC and spacer
  5'XXXXCNNNNNNNNCREADTCGTATGCCGTCTTCTGCTTG'3
     BC    UMI             SRA3 adapter
```

CLIPittyClip args:

```
     -u 8
     --bc-len 6
     --bc-first #to make it BC then UMI
     --spacer-len 1
     -a TCGTATGCCGTCTTCTGCTTG
```

3) iCLIP2 read architecture From Buchbender et al 2020 Methods. The Second UMI here is treated as a spacer. Args would be the same as for coCLIP with some light modifications.

```
  5'NNNNNXXXXXXNNNNREADTGAGATCGGAAGAGCGGTTCAG'3
     UMI1  BC  UMI2          L3 adapter
```

CLIPittyClip args:

```
     -u 5
     --bc-len 6
     --spacer-len 4
     -a TGAGATCGGAAGAGCGGTTCAG
```

Original iCLIP as in the 2010 NSMB paper has the following read architecture:

```
  5'XXNNNREADTGAGATCGGAAGAGCGGTTCAG'3
    BCUMI        L3 adapter
```

CLIPittyClip args in this case would be:

```
     -u 3
     --bc-len 2
     --bc-first #to make it BC then UMI
     --spacer-len 0
     -a TGAGATCGGAAGAGCGGTTCAG
```

4) iCLIP3 read architecture From Despic et al 2026 STAR Protocols. BC information is handled by illumina i5/i7 indices and is best done at the sequencing core facility stage. CLIPittyCLIP here assumes demultiplexed files with reads containing the following architecture.

```
  5'NNNNNNNNNREADNNNAGATCGGAAGAGCACACGTC'3
        UMI1     UMI2      L7 linker
```

CLIPittyClip args in this case would be:

```
     -u 9
     --trim3 3 #trims 3nt AFTER linker removal
     -a AGATCGGAAGAGCACACGTC
```

5) irCLIP and irCLIP2 read architectures The original irCLIP strategy (Zarnegar et al Nat Methods 2016 PMID: 27111506) is not dissimilar from iCLIP since it's only the 3' adapter that contains the ir label, and still uses circularization. irCLIP2 (Ducoli et al Nature 2025 PMID: 40140581) is more eCLIP style but with a twist: instead of 3' cDNA adapter ligation, cDNA reads are A- or C- tailed and used as a PCR handle.

irCLIP read architecture:

```
  5'NNNNNXXXXXXNNNWWWREADAGATCGGAAGAGCGGTTCAG'3
     UMI    BC     sp          Modified L3
```

CLIPittyClip args:

```
     -u 5
     --bc-len 6
     --spacer-len 6
     -a AGATCGGAAGAGCGGTTCAG
```

irCLIP2 read architecture:

```
  5'XXXXXXNNNNNNNNTTTTTTTTTTTTTREADAGATCGGAAGAGCACACGTC'3
      BC     UMI        sp               L7 linker
```

CLIPittyClip args:

```
     -u 8
     --bc-len 6
     --bc-first
     --spacer-len 12
     -a AGATCGGAAGAGCACACGTC
```

6) eCLIP read architecture (from Blue at al 2022 Nature Protocols PMID: 35322209 – see Fig S1). There are two versions, the original peCLIP and the newer seCLIP, which fixes the need to do paired end seq to get the UMI. Both use the following primers for making the final library with Illumina compatible barcodes for demultiplexing at the core facility stage.

D701 (read 2)
P7 adapter index read 1 i7-8mer multiplex R2 seq primer
5'CAAGCAGAAGACGGCATACGAGAT**XXXXXXXX**GTGACTGGAGTTCAGACGTGTGCTCTTCCGATC'3

D501 (read 1)
P5 adapter index read2 i5-8mer Truseq universal adapter
5'AATGATACGGCGACCACCGAGATCTACAC**NNNNNNNN**ACACTCTTTCCCTACACGACGCTCTTCCGATCT'3

For seCLIP the final library looks like this:

D501//ACACGACGCTCTTCCGATCT->3'
ACACGACGCTCTTCCGATCTNNNNNNNNNNREADAGAUCGGAAGAGCACACGUC
3'<-CTAGCCTTCTCGTGTGCAG//D701'5

So the final read for seCLIP is:

5'NNNNNNNNNNREADAGATCGGAAGAGCACACGTC'3
UMI InvRiL19 adapter

peCLIP has the UMI in the header so the read is simply:

5'READAGATCGGAAGAGCACACGTC'3
InvRiL19 adapter (aka L7 adapter)

CLIPittyCLIP args: --eclip se #or
--eclip pe

CLIPittyClip has specialized eCLIP modes. For se mode, UMI is hardcoded to length 10 and the adapter is similarly hardcoded to AGATCGGAAGAGCACACGTCTGAACTCCAGTCAC. For pe mode, UMI is already in the read header, so there's really just the 3' adapter trimming with a hardcoded eclip_adapters.fa file which contains all known eCLIP inline-barcode + TruSeq R2 sequences.

7) fPAR-CLIP read architecture (from Anastasakis et al 2017 NAR PMID: 33503264). PAR-CLIP in general uses old school sequential ligation of 3' and then 5' linkers, prior to RT. As such, truncations are not clonable by definition and so CITS will yield little if any results. While CLIPittyCLIP is not engineered to call T-to-C stats (see PCLIPtools Polash & Hafner NAR 2026 as the latest update to PARalyzer), the pipeline will output dedicated bed files for peaks or mutation types in both CTK and CLink modes.

fPAR-CLIP read architecture:

```
5'NNNNREADNNXXXXXXTGGAATTCTCGGGTGCCAAGG'3
   UMI    sp  BC         adapter
```

```
CLIPittyCLIP args:  --parclip
                    -u 4
```

For fPAR-CLIP, one only needs the length of the UMI (here 4nt), and 3' adapter trimming occurs with all hardcoded parclip_adapters.fa made from Table S1 in the 2017 NAR paper PMID: 33503264.

Legacy PAR-CLIP read architecture (from the original Hafner & Landthaler et al 2010 Cell paper PMID: 20371350, which references the contemporaneous 2010 Methods paper PMID: 18158127).

```
5'READTCGTATGCCGTCTTCTGCTTG'3
```

Many older PAR-CLIP (and early HITS-CLIP) datasets predate UMI and barcoding strategies, so standard CLIPittyCLIP args are all you need:

```
CLIPittyCLIP args:  --no-dedup #since there's no UMI
                    -a TCGTATGCCGTCTTCTGCTTG #original SRA3 linker
```

Extended notes on CLIP read architectures

iCLIP notes
(from 2010 NSMB Paper Konig et al). This is a step-by-step sanity check to make sure one understands how each of these protocols work.

L3: UGAGAUCGGAAGAGCGGUUCAG
RT primer: /5Phos/NNN**XX**AGATCGGAAGAGCGTCGTGGATCCTGAACCGCTC
(bold is BC, UMI)
Linearization primer: GTTCAGGATCCACGACGCTCTTCAAAA
Illumina 5′ end primer:
CAAGCAGAAGACGGCATACGAGATCGGTCTCGGCATTCCTGCTGAACCGCTCTTCCGATCT
Illumina 3′ end primer:
AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT

1)RT setup on L3 linker
  5′READUGAGAUCGGAAGAGCGGUUCAG′3
                <-3′CTCGCCAAGTCCTAGGTGCTGCGAGAAGGCTAGA**XX**NNN′5

2) Completed product
  5′READUGAGAUCGGAAGAGCGGUUCAG′3
  3′DAERACTCTAGCCTTCTCGCCAAGTCCTAGGTGCTGCGAGAAGGCTAGA**XX**NNN′5

3) Product with linearization primer after circ. BamHI site.
                       5′GTTCAGGATCCACGACGCTCTTCAAAA′3
                         ||||||||||||||||||||||||
  3′DAERACTCTAGCCTTCTCGCCAAGTCCTAGGTGCTGCGAGAAGGCTAGA**XX**NNNDAER′5
    |____|

4) Linearized cDNA
  3′GTGCTGCGAGAAGGCTAGA**XX**NNNDAERACTCTAGCCTTCTCGCCAAGTCCTAG′5

5) W/ illumina seq primers
  5′CAA//GAACCGCTCTTCCGATCT′3->
             ||||||||||||||
      3′GTGCTGCGAGAAGGCTAGA**XX**NNNDAERACTCTAGCCTTCTCGCCAAGTCCTAG′5
                                        ||||||||||||||
                                  <-3′TCTAGCCTTCTCGCAGCAC//TAA′5
Final read from sequencer:
  5′**XX**NNNREADTGAGATCGGAAGAGCGGTTCAG′3
    BCUMI   L3 adapter

iCLIP2 read architecture (from Buchbender 2020 Methods Fig 2) — similar idea, but now with UMI before the BC:
  5′NNNNN**XXXXXX**NNNNREADTGAGATCGGAAGAGCGGTTCAG′3
    UMI1   BC UMI2   L3 adapter

iCLIP3 notes
(from 2026 STAR Protocols, Despic et al). Here circularization is replaced with a 3' linker ligation step on cDNA that introduces the UMI after RT. P5 and P7 PCR handles make this compatible with illumina dual indexing for barcodes.

L7 linker:/5rApp/NNNAGATCGGAAGAGCACACGTC/3ddC/
RT primer: GACGTGTGCTCTTCC
Second adapter:/5Phos/NNNNNNNNNAGATCGGAAGAGCGTCGTG/3ddC/
P5 short (P5S): ACACGACGCTCTTCCGATC*T
P7 short (P7S): GACGTGTGCTCTTCCGATC*T
P5 long (P5L):
AATGATACGGCGACCACCGAGATCTACAC**XXXXXXXX**ACACTCTTTCCCTACACGACGCTCTTCCGATC*T
P7 long (P7L):
CAAGCAGAAGACGGCATACGAGAT**XXXXXXXX**GTGACTGGAGTTCAGACGTGTGCTCTTCCGATC*T

1)RT setup on L7 linker
5'READNNNAGATCGGAAGAGCACACGTC'3
<-CCTTCTCGTGTGCAG RT primer

2) Completed product with second adapter ligation setup

2nd adapter RT product
3'GTGCTGCGAGAAGGCTAGANNNNNNNNN'5 + 3'DAERNNNTCTAGCCTTCTCGTGTGCAG'5

3) Short PCR setup

P5S primer
5'ACACGACGCTCTTCCGATCT ->
3'GTGCTGCGAGAAGGCTAGANNNNNNNNNDAERNNNTCTAGCCTTCTCGTGTGCAG'5
<-TCTAGCCTTCTCGTGTGCAG'5
P7S primer

4) Long PCR setup same as step 3 with longer dual index primers

Final read from sequencer:

5'NNNNNNNNNREADNNNAGATCGGAAGAGCACACGTC'3
UMI1 UMI2 L7 linker

BrdU-CLIP notes

From Weyn-Vanhentenryck 2014 Cell Rep PMID: 24613350. Note that primers initially had 4mer barcodes, plus an anchor position C at either end of the UMI. Will treat the first anchor as part of the BC and the second anchor as part of the UMI.

SRA3: /5Phos/UCGUAUGCCGUCUUCUGCUUG-puromycin-3 OR
Pre-A SRA3: /5rApp/TCGTATGCCGTCTTCTGCTTG/3ddc/-3'
RT-1 primer:
5'pGNNNNNNNNG**XXXX**GATCGTCGGACTGTAGAACTCT/idSp/CAAGCAGAAGACGGCATACGA
P5 primer (fwd):
AATGATACGGCGACCACCGACAGGTTCAGAGTTCTACAGTCCGACG
P7 primer (rev):
CAAGCAGAAGACGGCATA
Illumina Small RNA Sequencing Primer:
CGACAGGTTCAGAGTTCTACAGTCCGACGATC

1)RT setup on L2 primer
5'READUCGUAUGCCGUCUUCUGCUUG'3
3'AGCATACGGCAGAAGACGAAC/idSp/TCTCAAGATGTCAGGCTGCTAG**XXXX**GNN//G'5

2) Completed product, circularized
5'READUCGUAUGCCGUCUUCUGCUUG'3
DAERAGCATACGGCAGAAGACGAAC/idSp/TCTCAAGATGTCAGGCTGCTAG**XXXX**GNN//G'5

3) Final read w/ PCR primers:

P5///TTCAGAGTTCTACAGTCCGACG->
ACAGGTTCAGAGTTCTACAGTCCGACG**XXXXC**NNNNNNNNCREADTCGTATGCCGTCTTCTGCTTG
<-ATACGGCAGAAGACGAAC

Final read from sequencer:

C C Anchor positions, treated as BC and UMI
5'**XXXXC**NNNNNNNNCREADTCGTATGCCGTCTTCTGCTTG'3
BC UMI SRA3 adapter

CoCLIP cloning notes

Current lab standard as published in Yi et al RNA 2024, derived from Darnell and Zhang labs BrdU-CLIP protocols that were first published in the 2014 Cell Reports paper (PMID: 24613350). The RT primers in our implementation were first described in Sawicka et al 2019 eLife (PMID: 31860442)

L32 linker: /5rApp/GTGTCAGTCACTTCCAGCGG/3ddC/

RT primer

/5Phos/DDD**XXXXXX**NNNNNNNAGATCGGAAGAGCGTCGT/iSp18/CACTCA/iSp18/CCGCTGGAA GTGACTGAC

DP5-PE AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT

SP3-PE CAAGCAGAAGACGGCATACGAGATCTCGGCATTCCTGCCGCTGGAAGTGACTGACAC

1) RT step with barcoded UMI-containing primer

READGTGTCAGTCACTTCCAGCGG

<-CAGTCAGTGAAGGTCGCC[ ]ACTCAC[ ]TGCTGCGAGAAGGCTAGANNNNNNN**XXXXXX**DDD

2) After circularization:

UMI BC sp L32 adapter

5'ACGACGCTCTTCCGATCTNNNNNNN**XXXXXX**DDDREADGTGTCAGTCACTTCCAGCGG'3

3) PCR

DP5-PE ACGACGCTCTTCCGATCT->

5'ACGACGCTCTTCCGATCTNNNNNNN**XXXXXX**DDDREADGTGTCAGTCACTTCCAGCGG'3

<-CACAGTCAGTGAAGGTCGCCSP3

irCLIP notes

3′ adapter (from original irCLIP Nat Methods 2016)
L3-azide-biotin:
AGATCGGAAGAGCGGTTCAGAAAAAAAAAAAA/iAzideN/AAAAAAAAAAAA/3Bio/-3′

RT primer - this is a circularization iCLIP style library prep
/5phos/WWW.NNN**XXXXXX**NNNNNTACCCTTCGCTTCACACACAAG/iSp18/GGATCC/iSp18/TAC
TGAACCGC
<-CGCCAAGTCAT...RT primer handle
5′NNNNN**XXXXXX**NNNWWWREADAGATCGGAAGAGCGGTTCAG
UMI BC sp L3 mod (original L3 starts GTAGA)

irCLIP2 notes

3′ adapter (from irCLIP-RNP and reclip paper Nature 2025) – this is more eCLIP style with a twist: instead of 3′ cDNA adapter LIGATION, cDNA reads are tailed and used as a PCR handle.

5′-/5phos/AGATCGGAAGAGCACACGTCTGAACTCCAGTCAC/Barcode
6nt/ATCTCGTATGCCGTCTAAAAAAAAAAAAAAAAAAA/iAzideN/CT/3bio/-3′
P7-end cDNA primer: 5′-CAAGCAGAAGACGGCATACGAGAT-3′
Annealed-barcoded-2nd-strand forward 1 for A-tails:
5′-TCCCTACACGACGCTCTTCCGATCT**XXXXX**NNNNNNNNTTTTTTTTTTTTVN-3′
Annealed-barcoded-2nd-strand forward 2 for C-tails:
5′-TCCCTACACGACGCTCTTCCGATCT**XXXXX**NNNNNNNNGGGGGGGGHN-3′
Annealed-barcoded-2nd-strand reverse: 5′-TCAGTAGATCGGAAGAGCGT/3ddC/-3′
P5-PCR1 A: 5′-CGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT-3′
P5-PCR1 C: 5′-CCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT-3′
P5-Solexa
5′-AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT-3′

1)RT setup with P7-end cDNA primer
5′READAGATCGGAAGAGCACACGTCTGAACTCCAGTCAC**XXXXXX**ATCTCGTATGCCGTCTAA//′3
||||||||||||||||
P7 <-TAGAGCATACGGCAGAAGAC//

2)Completed product with 3′ A tailing, 2nd strand fwd 1
NNNNNNNNTTTTTTTTTTTTVN-> P5-PCR1 A
3′AAAAAAAAAAAADAERTCTAGCCTTCTCGTGTGCAGACTTGAGGTCAGTG**XXXXXX**//′5

3) P5 PCR1 and P7 end cDNA amplification
//GAGATACACGACGCTCTTCCGATCT-> P5
//TCCCTACACGACGCTCTTCCGATCT**XXXXX**NNNNNNNNTTTTTTTTTTTTREAD//**XXXXXX**ATCTC
P7 <-TAGAG
Read architecture (irCLIP2):
5′**XXXXXX**NNNNNNNNTTTTTTTTTTTTREADAGATCGGAAGAGCACACGTC...′3
UMI BC sp L7 linker

eCLIP notes

From Blue et al 2022 Nat Protocol PMID: 35322209. See Fig S1.

D701 (read 2)
P7 adapter index read 1 i7-8mer multiplex R2 seq primer
5'CAAGCAGAAGACGGCATACGAGAT**XXXXXXXX**GTGACTGGAGTTCAGACGTGTGCTCTTCCGATC’3

D501 (read 1)
P5 adapter index read2 i5-8mer Truseq universal adapter
5'AATGATACGGCGACCACCGAGATCTACAC**NNNNNNNN**ACACTCTTTCCCTACACGACGCTCTTCCGAT
CT’3

InvRiL19 (as RNA, 3’ adapter for CLIPed RNA)
AGAUCGGAAGAGCACACGUC/3SpC3/
<-AGCCTTCTCGTGTGCAGAC

InvAR17 (RT primer off of InvRiL19)
CAGACGTGTGCTCTTCCGA

InvRand3Tr3 (3’ adapter for cDNA)
NNNNNNNNNNAGATCGGAAGAGCGTCGTGT/ 3SpC3/

For seCLIP the final library looks like this:

D501//ACACGACGCTCTTCCGATCT->3’
ACACGACGCTCTTCCGATCTNNNNNNNNNNREADAGAUCGGAAGAGCACACGUC
3’<-CTAGCCTTCTCGTGTGCAG//D701’5

So the final read for seCLIP is:

5’NNNNNNNNNNREADAGAUCGGAAGAGCACACGUC’3
UMI InvRiL19 adapter

peCLIP has the UMI in the header so the read is simply:

5’READAGAUCGGAAGAGCACACGUC’3
InvRiL19 adapter

fPAR-CLIP read notes (from Anastasakis et al 2017 NAR Fig S1 PMID: 33503264)

Fluorescent barcoded 3'adapter (BC in bold):
rAppNN**XXXXXX**TGGAATTCTCGGGT(fl)GCCAAGG-fl
5'adapter:
(aminolinker-)GTTCAGAGTTCTACAGTCCGACGATCrNrNrNrN
RT primer: GCCTTGGCACCCGAGAATTCCA
5' PCR Short: CTTCAGAGTTCTACAGTCCGACGA
5' long: AATGATACGGCGACCACCGAGATCTACACGTTCAGAGTTCTACAGTCCGA
3'IndexPrimer (BC in bold):
CAAGCAGAAGACGGCATACGAGAT**XXXXXX**GTGACTGGAGTTCCTTGGCACCCGAGAATTCCA

1st short PCR: cDNA after sequential 3' and 5' linker ligation, w/ RT primer and 5' PCR short primer. Note extra nucleotide at 3' end added and the C in the 5' short PCR primer that helps avoid PCR artifacts.

```
CTTCAGAGTTCTACAGTCCGACGA->
 ||||||||||||||||||||||||
GTTCAGAGTTCTACAGTCCGACGATCNNNNREADNNXXXXXXTGGAATTCTCGGGTGCCAAGG
                                            ||||||||||||||||||||
                                          <-ACCTTAAGAGCCCACGGTTCCG
```

2nd PCR

```
AAT//GTTCAGAGTTCTACAGTCCGA->
      ||||||||||||||||||||
     CTTCAGAGTTCTACAGTCCGACGATCNNNNREADNNXXXXXXTGGAATTCTCGGGTGCCAAGGC
                                                ||||||||||||||||||||
                                              <-ACCTTAAGAGCCCACGGTTCCT/
```

For the 3'linker the XXXXXX barcode can be used to label samples (i.e. cytoplasmic or nuclear CLIPs as in the PKM2 PAR-CLIP paper as a guide). As far as I can tell, these linkers add BC information at the ligation step, but are not used as primary de-multiplexing barcodes.

NN**TAGTCG**TGGAATTCTCGGGTGCCAAGG for cytoplasm
NN**AGTGTC**TGGAATTCTCGGGTGCCAAGG for nuclear
43 additional barcoded 3' linkers in Table S1 of the fPAR-CLIP NAR paper

Full product:

```
                        Illumina v1.5 Small RNA-Seq Read 1
                            ACACGTTCAGAGTTCTACAGTCCGACGATC->
5'AATGATACGGCGACCACCGAGATCTACACGTTCAGAGTTCTACAGTCCGACGATCNNNNREADNNXXX
XXXTGGAATTCTCGGGTGCCAAGGAACTCCAGTCAC**XXXXXX**ATCTCGTATGCCGTCTTCTGCTTG'3
   TGGAATTCTCGGGTGCCAAGGAACTCCAGTCAC->
         Lexogen 3' adapter i7
```